

IC-32 R-I-V IN SERIES, PARALLEL AND COMBINATION CIRCUITS

9/12/2023

FOR IN-CLASS AND TAKE-HOME LAB KIT

32-1 OBJECTIVE

The purpose of this lab is to experiment with the variables that contribute to the operation of an electrical circuit, specifically the resistance, voltage and current in series circuits, parallel circuits and series-parallel circuits.

32-3 EQUIPMENT

AC/DC Electronics Lab Board

Assorted Resistors

Multimeter, Voltage and Current Sensor

One or two D-cells

Wires

Capstone Software



32-3 THEORY

In this Lab we will study the effects on the values of the net resistance, voltage drops and current when resistors are placed in series, parallel or series-parallel combination with each other. These circuits are shown in Figure-1.

When resistances are placed in series or parallel, the total, or equivalent, resistance is given by the following equations:

Series: $R_{eq} = R_1 + R_2 + R_3 + \dots$ (1)

Parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$ (2)

The Current through one or a combination of resistors is given by Ohm's Law:

$$I = \frac{\Delta V}{R} \quad (3)$$

We will measure the voltage across the resistors and the current through the resistors and compare these to the calculated values.

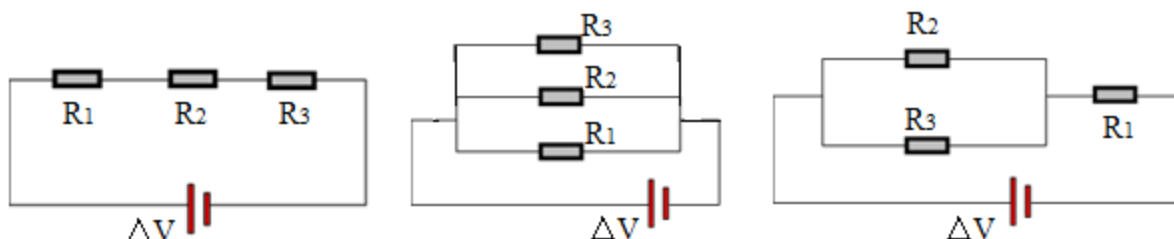


Figure-1: Circuits showing resistors in Series, Parallel and Series-Parallel Circuits.

The resistors are labelled by colour coding shown in Figure 2.

Figure 2: Color Coding for Resistors

<u>First Three Bands</u>						<u>Fourth Band</u>	
Black	0	1st digit	2nd digit	Number of zeroes	Tolerance	None	± 20%
Brown	1					Silver	±10%
Red	2					Gold	± 5%
Orange	3					Red	± 2%
Yellow	4						
Green	5						
Blue	6						
Violet	7						
Gray	8						
White	9						

ex: Orange - Green - Red - Gold				
3	5	2	5%	= 3500 ± 5% Ω

32-4 PROCEDURE

To measure the resistance, you will need the Multimeter. However, to measure the voltages and currents you may use either the Multimeter or the Voltage and Current Sensors, depending on what is available to you. In this Manual, we will be using the terms “Voltage Sensor” and “Current Sensor” to mean the device used to measure these parameters.

Choose three resistors from those available. They may be of same or different values, but should range between 100 Ω to 1000 Ω (i.e. not smaller than 100 Ω and not larger than 1000 Ω). Refer to them as resistor #1, #2 and #3. You need to remember or label them so that they don't get mixed up.

Determine the color code of your resistors. Enter the Colors, Coded Value and Tolerance as indicated by the color of the fourth band. Use the Multimeter to measure the resistance of each of your three resistors. Determine the percentage experimental error of each resistance value and enter the values in the appropriate columns. Use the measured values in all subsequent work.

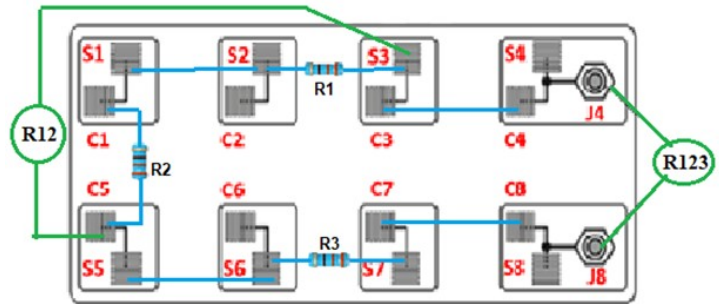
1-SERIES CIRCUIT

Resistances in Series

1. Connect the three resistors into the Series Circuit shown in Fig. 3, using the spring clips on the Circuits Experiment Board. DO NOT attach the battery to the circuit. Measure the resistances R12, R23 and R123 and note in Table-1. The resistances are measured by connecting the leads of the Multimeter at the points across which the resistance is required. In Fig. 3 the placement of prongs of the meter for measuring R12 and R123 are shown in green color. Enter the measured and calculated values,

and the percent error between these. Use the coded value as the theoretical value in finding the percent error. Use the measured value in all subsequent calculations.

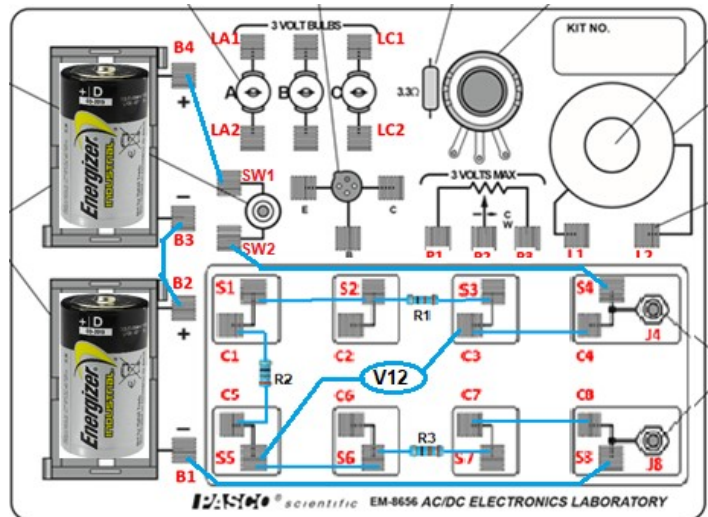
Figure-3: Placement of resistors and Ohm-meter prongs to measure resistances.



Voltages in Series

2. In the circuit already made, add two D-Cells. Connect B2 and B3 so that the two batteries are now in series with each other.
3. Connect B4 to SW1, SW2 to S4, and S8 to B1. The circuit should now be as in Figure 4.
4. When the switch is pressed, the current will flow from the battery in the direction: B4 - SW1 - SW2 - S4 - S3 - R1 - S2 - S1 - R2 - C5 - C6 - R3 - S7 - S8 - B1.
5. Connect the Voltage Sensor to Capstone (or use the multimeter). First Tare it, and then use it to measure voltages V1, V2, V3, V12, V23 and V123 and note in Table 1. To measure the voltage across any resistor, place one prong of the voltage sensor at one side of the resistance, and the other prong on the other side. For measuring, e.g. V12, one prong is on one side of R1 and the other is on the far side of R2. (See Fig. 4). Current will not flow through Voltage sensor as it has very high resistance.

Figure-4: Placement of resistors and volt-meter prongs to measure voltage.



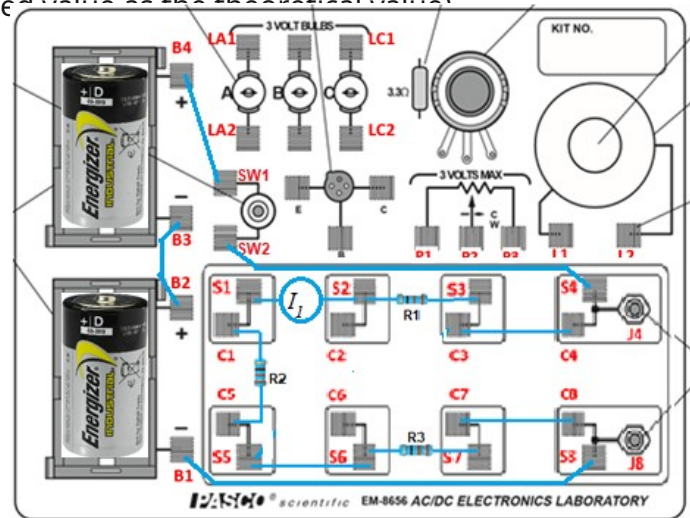
Currents in Series

6. Connect the Current Sensor to Capstone Software (or use the Multimeter).
7. In the circuit already made, remove the wire connecting C4 to C3. Insert the Current Sensor between C3 and C4 by connecting its one end to C3 and the other end to C4. This will measure I_o . The current is flowing from C4 towards C3. Make sure that the

polarity of the sensor is correct. Also, start with a high current range, and reduce the max range as needed.

8. Press Record in Capstone, press the switch for a few seconds, release the switch, stop recording in Capstone. Note the current value in Table 1.
9. Remove the current sensor and replace the wire between C3 and C4.
10. Repeat the process on procedures 7 to 9 for the wires between S1 and S2 (I_1 as shown in Fig 8), between S5 and S6 (I_2), and between S7 and S8 (I_3). Note the values of currents in Table 1.
11. Calculate and note what the current should be based on Ohm's Law, i.e. $I = V / R$ where R and V are the resistance and voltage between the two points. Calculate and note the percent error (use the measured value as the theoretical value)

Figure-5: Placement of resistors and Current Sensor prongs to measure current I_1 .



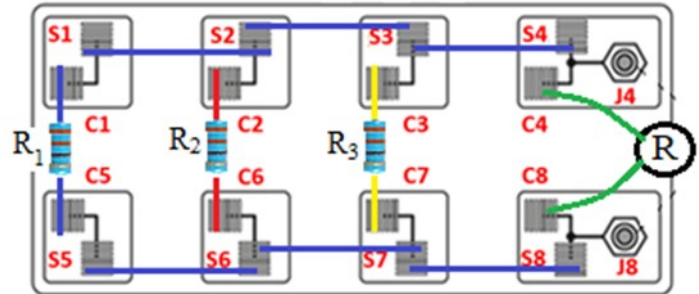
2- PARALLEL CURCUIT

Resistances

1. In the existing circuit, leave the battery and switch connections to S4 and S8. Take out the three resistors and re-attach them as shown in Fig. 6 (only a part of the board is shown). Do not press the switch while taking the resistance measurements.
2. Connect the Multimeter across C4 and C8. Set it up for resistance measurement.
3. In Fig. 6, the value of resistance that the Multimeter is showing is that of R123, i.e. R1, R2 and R3 connected in parallel. Note this value in Table 2A as the measured value of R123.
4. Now take out resistance R_3 from the circuit. To do this, you may disconnect any one side of the yellow-colored wire, i.e. at C3 or C7, or both sides. Now measure the resistances across C4 and C8 in the already connected Multimeter. This will be the resistance across R1 and R2 connected in parallel. Note this in Table 2A as the measured resistance for R12.

5. To measure R_{13} , reconnect R_3 and take out R_2 (the red wire).
6. To measure R_{23} , reconnect R_2 and take out R_1 (the blue wire).
7. Use the measured values of R_1 , R_2 and R_3 from Table 1, to obtain the calculated values of R_{12} , R_{23} , R_{13} , and R_{123} , as well as the percent errors.

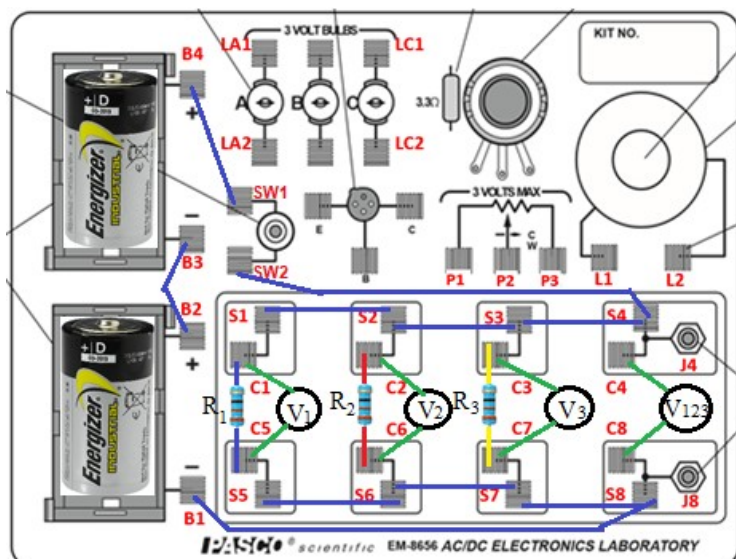
Figure-6: Diagram for Resistors in Parallel and Ohm-meter prongs to measure R_{123} in parallel.



Voltages

8. Remove the Multimeter and connect the Voltage Sensor to Capstone.
9. When the switch is pressed, the current will flow from the battery in the direction: B4 - SW1 - SW2 - S4 - S3 - S2 - S1. Then through R_1 , R_2 and R_3 to S5, S6 and S7 and then to S8 - B1. This completes the circuit.
10. Use the Voltage Sensor to measure the voltages V_1 , V_2 , V_3 and V_{123} across R_1 , R_2 , R_3 and R_{123} respectively, by connecting the Voltage Sensor at the points as shown by the green lines in Fig-7 (of course, only one at a time, if you have only one voltage sensor). Record in Table 2B.

Figure 7: Diagram for Resistors in Parallel and Volt-meter placements.



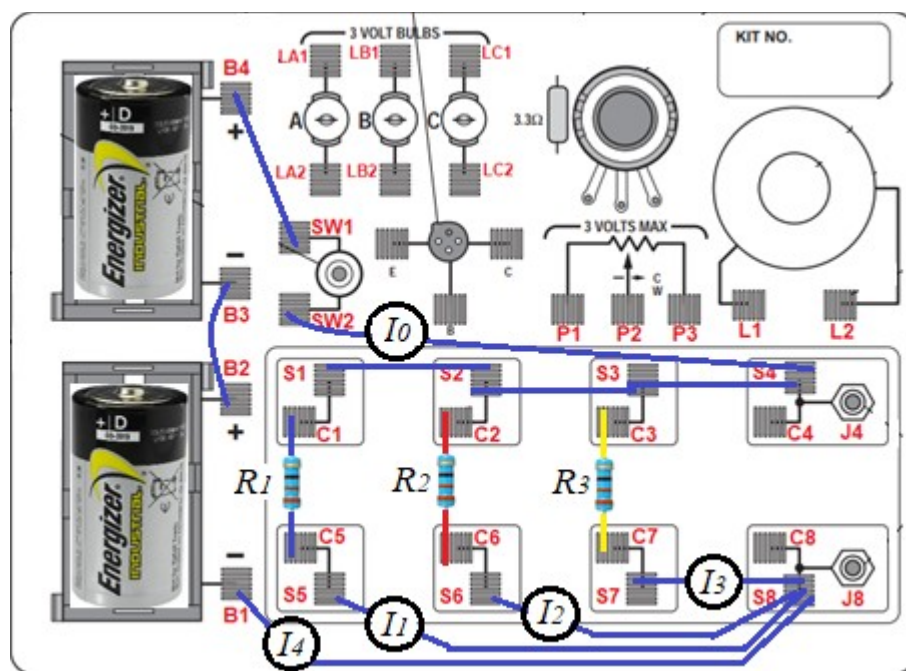
Currents

11. Now remove the wires between S5 and S6, and between S6 and S7. Connect S5 and S6 directly to S8 (see Fig. 8). To measure the current, you will connect the Current Sensor in the circuit at different positions. For this you will have to remove the connecting wires and replace them with the current sensor.
12. Connect the Current Sensor to Capstone Software.
13. First, remove the wire connecting the switch SW2 to S4, and attach the Current Sensor in its place. Make sure that the polarity of the sensor is correct. Press Record in Capstone, press the switch for a few seconds, release the switch, stop recording in Capstone. Note the current value I_0 in Table 2B. Remember to start with a high current range and reduce the max range as needed.
14. Now replace the wire between SW2 and S4. Remove the wire between S5 and S8 and replace it with the Current Sensor. Measure and note the current value I_1 in Table 2B.
15. Replace the wire between R1 and S8 and repeat the process for R2 and R3. Obtain I_2 and I_3 and record them.
16. Lastly, replace the wire between S8 and B1 with the current sensor, and measure I_4 .
17. Calculate and note what the current should be based on Ohm's Law, i.e.

$$I = V / R$$

Where R and V are the resistance and voltage between the two points. Calculate and note the percent error (use the measured value as the theoretical value).

Figure 8: Diagram for Resistors in Parallel and Ammeter placements.



3-SERIES-PARALLEL COMBINATION CIRCUIT

Resistances

1. Make the circuit as shown in Figure 9.
2. Measure the resistances R_1 , R_2 and R_{123} and record them in Table 3A. The positions of the Multimeter for these are the same as those shown for the Voltmeter in Fig-9.

Voltages

3. Measure the voltages V_1 , V_2 (across R_2), V_3 (across R_3), V_{23} (across R_{23}) and V_{123} , and record them in Table 3B.

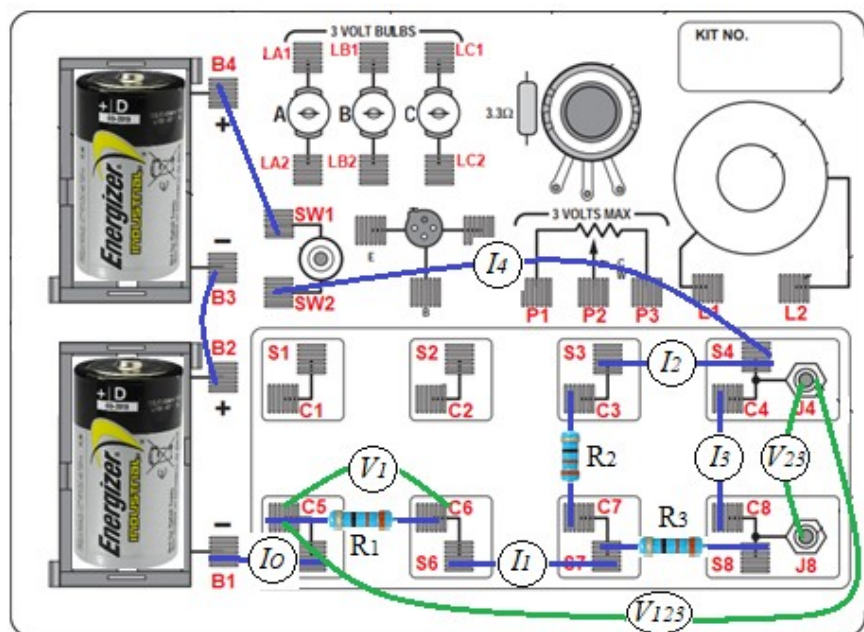
Currents

4. Measure the currents I_0 , I_1 , I_2 , I_3 and I_4 at locations shown in the figure and record them. Remember that you have to remove the wire connection between these points and replace them with the current sensor.
5. Calculate and note in Table 3B what the current should be based on Ohm's Law, i.e.

$$I = V/R$$

Where R and V are the resistance and voltage between the two points. Calculate and note the percent error (use the measured value as the theoretical value).

Figure 9: Diagram for Resistors in Series-Parallel Combination and Current and Voltage Sensor placements



32-5 PRECAUTIONS

1. Label the resistors carefully. Make sure that you do not mix them up.
2. Start measurements at a higher range and progressively lower the range in the instruments, especially the current sensor / ammeter.
3. After every current measurement, be sure to replace the wires to re-connect the two points between which the current was measured.

32-6 DATA SHEET: RESISTORS, VOLTAGES AND CURRENTS IN CIRCUITS

RESISTANCES

Values of resistances from Color coding and measurements

No	Colors 1 st 2 nd 3 rd 4 th	Coded Resistance	Tolerance	Measured Resistance	Percent error
R1					
R2					
R3					

SERIES CIRCUIT

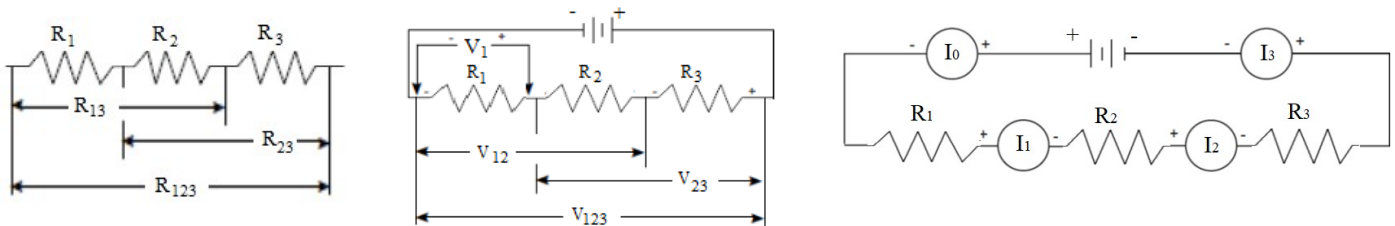


Table 1: Measured values of Resistances, Voltages and Currents, and calculated currents and percent errors.

	Measured Resistance		Measured voltage		Measured current	Calculated Current	% ERROR
Units		Units		Units			
R ₁		V ₁		I ₀			
R ₂		V ₂		I ₁			
R ₃		V ₃		I ₂			
R ₁₂		V ₁₂		I ₃			
R ₂₃		V ₂₃					
R ₁₂₃		V ₁₂₃					

PARALLEL

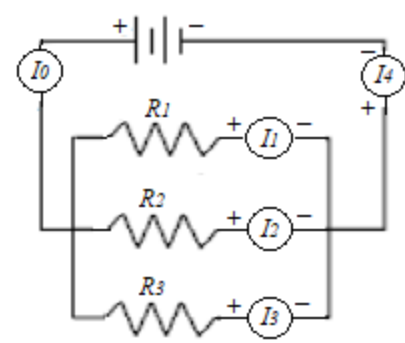
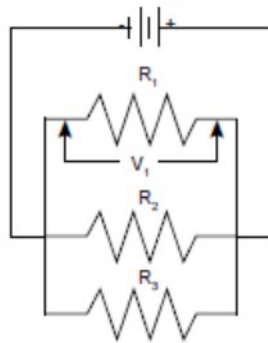
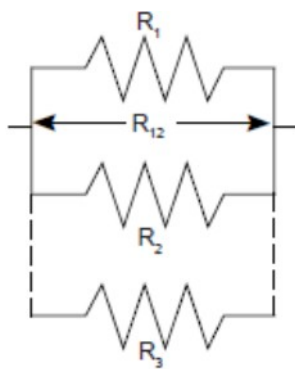


Table 2A: Measured values of Resistances in Parallel.

	Measured resistance	Calculated resistance	Percent error
R12			
R13			
R23			
R123			

Table 2B: Measured values of Voltages and Currents across Resistors in Parallel

	Measured Resistance from Tables 1 and 2A		Measured voltage		Measured current	Calculated Current	% Error
Units		Units		Units			
R ₁		V ₁		I ₁			
R ₂		V ₂		I ₂			
R ₃		V ₃		I ₃			
R ₁₂₃		V ₁₂₃		I ₄			

				I_0			
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SERIES-PARALLEL COMBINATION CIRCUITS

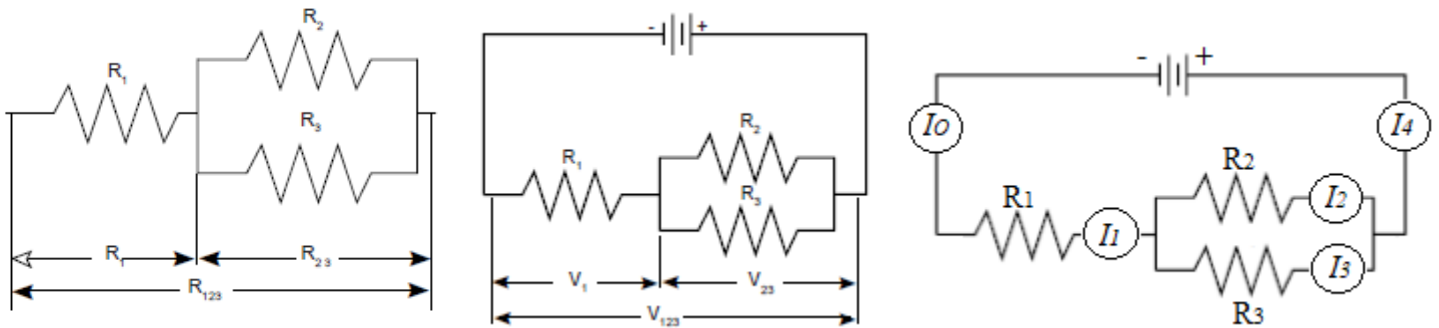


Table 3A: Measured values of Resistances in Combination.

	Measured resistance	Calculated resistance	Percent error
R1			
R23			
R123			

Table 3B: Measured values of Voltages and Currents across Resistors in Combination

	Measured Resistance from Table 3A		Measured voltage		Measured current	Calculated Current	% Error
Units		Units		Units			
R_1		V_1		I_1			
R_{23}		V_2		I_2			
R_{123}		V_3		I_3			
		V_{23}		I_4			

	V_{123}		I_0			
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32-7 REPORT:

Upload the following in the Report for this Lab.

		Points
1.	Data Tables for Series Circuits	10
2.	Data Tables for Parallel Circuits	10
3.	Data Tables for Combination Circuits	10
4.	Sample Calculations	5
5.	Sources of Error in this experiment.	5
6.	Discussion of Results	10
	Total	50

32-8 ADDITIONAL INFORMATION

Hewitt Drew It 96: Equivalent Resistance

http://www.hewittdrewit.com/Links/96_Equivalent_Resistance%C2%A0.html

PhET Simulation Circuit Construction Kit DC:

https://phet.colorado.edu/sims/html/circuit-construction-kit-dc-virtual-lab/latest/circuit-construction-kit-dc-virtual-lab_all.html

32-9 POINTS TO THINK ABOUT

1. Why does the resistance decrease when resistances are connected in parallel?
2. Why does the resistance increase when resistances are connected in series?
3. Why are the voltage drops across R_1 , R_2 and R_3 the same in the case of parallel circuit?

32-10 SAMPLE DATA / GRAPHS